

# Yuxian Li

liyuxian1358@gmail.com Website LinkedIn GitHub

Interested in how to steer AI to continuously learn from messy, heterogeneous physical-world data, and designing the low-level compute infrastructure to scale it.

## EDUCATION

**The University of Hong Kong**

B.Eng. in Computer Engineering



Expected 2027

Selected Courses: Probability & Statistics, Linear Algebra, Discrete Math, Computer Architecture, Quantum Computing  
**Outstanding Poster Award**, HKU ECE Poster Display Day, 2026 (1 of 13 recipients in a 158-student graduating cohort)

**University of California, Berkeley**

Visiting Student. Data Structures and Algorithms

Summer 2023

## RESEARCH EXPERIENCE

**CAFE: Counterfactual Benchmark for Segmentation Models**<sup>[1]</sup> [arXiv](#) [Project Page](#)

2026

Agent Systems and Evaluation Lead, mentored by Profs. Xihui Liu (HKU MMLab) and Han Wang

- Even strong segmentation models can draw the right outline but identify the wrong object or concept. So I discussed these issues with my coauthors to formulate the research problem, and together we built CAFE to expose and quantify them.
- Co-curated 2,146 cases from 48,423 AI-edited candidates through three-stage expert review, checking edit quality while retaining the target's location and outline. We evaluated six segmentation systems (e.g., end-to-end, modular, and agentic) with correct and misleading prompts across 4,292 episodes.
- Conceived and built CAFE-SAM3, a GPT-5.5 visual agent that revises prompts and zooms in on material and texture cues to select a matching mask or abstain. The agent raised class-gated F1 from 38.5 to 63.3 (+24.8 points) and reduced target-aligned concept swaps from 8.9% to 1.7% on CAFE.
- Led review-stage follow-up experiments and failure analysis to address reviewer questions. I built a fault-tolerant Ray evaluation pipeline with GPU-resident SAM 3 actors, checkpoint recovery, and traceable reruns to support these studies.

**Human-Signal-Guided Representation Learning for VLA Policies**

2026

Research Intern at [HKU InfoBodied AI Lab](#), mentored by Prof. Yanchao Yang

Manuscript in preparation.

- Co-developing human-signal-guided representations for robot policy learning. I contribute to temporal alignment across modalities and design learning objectives and ablation studies.

**Continuous-Time ODE Solver for Flow-Matching Action Generation**

2025

Research Assistant, Center for Advanced Semiconductors and Integrated Circuits (CASIC), HKU

Supervised by Prof. Han Wang. Manuscript under review.

- On-device inference latency limits how quickly learned robot policies can respond in dynamic environments. I designed circuit-compatible flow-matching action models for faster closed-loop control.
- Designed the distillation method and built training and evaluation pipelines, with ablations across four [LIBERO](#) suites and real-robot tasks.
- Designed three language-conditioned tasks and built the robot-camera setup, including custom 3D-printed fixtures and teleoperated data collection.
- Collaborated closely with circuit and chip designers on model-hardware co-design, including noise- and quantization-aware training.

## PUBLICATIONS

<sup>[1]</sup> **From Pixels to Concepts: Do Segmentation Models Understand What They Segment?**

S. Liang\*, Z. Wang\*, Y. Li\*, X. Liu, and H. Wang

*NeurIPS 2026 Evaluations & Datasets Track (Poster)*

\*Equal contribution (co-first authors).

INDUSTRY EXPERIENCE

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| <b>OneRobotics (HKEX: 6600)</b><br><i>Research Intern, Multimodal Foundation Models</i>                                                                                                                                                                                                                                                                                                                                                                                                         | Aug to Sep 2026<br>Shenzhen, China |
| <ul style="list-style-type: none"><li>Driving online RL for humanoid loco-manipulation with VLA and world-action models (WAMs) for reliable household deployment. I am co-developing a closed-loop RL pipeline that uses rollout feedback to update and redeploy policies.</li><li>Developing and scaling multimodal pretraining pipelines for VLA and WAMs. I work with colleagues to refine episode/subtask labeling, cross-embodiment retargeting, and UMI/egocentric data mixing.</li></ul> |                                    |
| <b>Inspur Group</b><br><i>Software Development Intern</i>                                                                                                                                                                                                                                                                                                                                                                                                                                       | Sep to Dec 2024<br>Jinan, China    |
| <ul style="list-style-type: none"><li>Built front-end workflows for an enterprise platform and worked with backend engineers to integrate and test APIs against business requirements.</li></ul>                                                                                                                                                                                                                                                                                                |                                    |

OPEN SOURCE

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| <b>Scholar-MCP</b> <a href="#">GitHub</a><br><i>Lead Developer and Maintainer</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2025 to Present |
| <ul style="list-style-type: none"><li>An MCP server for agentic auto-research that helps agents find papers and reuse findings. I built parallel multi-source search and reranking to improve retrieval efficiency and limit source bias, raising top-five query hit rate from 32% to 62% on a fixed 50-query <a href="#">LitSearch</a> sample.</li><li>Designed tools for agents to refine queries and follow citations, with source-linked, length-bounded responses to steer long-running research. A searchable SQLite-backed library preserves papers and notes across sessions, reducing repeated searches and context overhead.</li></ul>                                                                                   |                 |
| <b>LeRobot</b> 🤖 <a href="#">GitHub</a><br><i>Upstream Contributor, Hugging Face</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2026            |
| <ul style="list-style-type: none"><li>RealSense D405 startup failures previously required a hard manual USB reset, so I added automatic recovery in merged <a href="#">PR #3894</a> with thread-safe frame handling and exception-safe cleanup to prevent stale-frame writes during restarts and improve startup reliability.</li><li>To make robot policies more robust to real-world visual variation, I implemented configurable PyTorch augmentations and integrated them into LeRobot's torchvision v2 pipeline via <a href="#">PR #4210</a> and released <a href="#">robo-augment</a> as a reusable library.</li></ul>                                                                                                       |                 |
| <b>Reveal StarMap</b> <a href="#">GitHub</a> <a href="#">Steam Workshop</a><br><i>Lead Developer and Maintainer</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2026            |
| <ul style="list-style-type: none"><li>A C# mod for the game Oxygen Not Included that lets players preview the full star map without changing saved exploration progress. I reverse-engineered Unity's map and camera behavior and used Harmony patches to preserve and restore exploration and camera state.</li><li>Handled camera bounds across scene changes and save loading, and restored state after mod conflicts. I published it on Steam Workshop, gaining thousands of subscribers.</li></ul>                                                                                                                                                                                                                            |                 |
| <b>MediBot</b> <a href="#">GitHub</a><br><i>Project Lead (3-person team)</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2024 to 2025    |
| <ul style="list-style-type: none"><li>Led end-to-end development of an autonomous mobile robot for contactless hospital patrols during COVID-19, integrating a Jetson Nano and mecanum base with Arduino motor control and ESP32 telemetry.</li><li>To support autonomous patrol, I published encoder odometry and TF in ROS and configured YDLidar navigation with GMapping, AMCL localization, NavFn global planning, DWA local planning, and layered costmaps, producing a lab map with 5 cm resolution.</li><li>Integrated thermal imaging and camera-based face and mask screening. I built a remote interface with live camera and LiDAR views, MQTT telemetry and motion commands, and emergency manual takeover.</li></ul> |                 |

SELECTED MINI-PROJECTS

|                                                                                                                                  |              |
|----------------------------------------------------------------------------------------------------------------------------------|--------------|
| <b>Online Mental Health Risk Predictor</b> <a href="#">GitHub</a><br><i>Project Lead (3-person team)</i>                         | 2026         |
| <b>Parallel Jacobi Solver</b> <a href="#">GitHub</a>                                                                             | 2025         |
| <b>UniHaven Off-Campus Housing Platform</b> <a href="#">GitHub</a><br><i>Requirements and Backend Engineer (4-person team)</i>   | 2024 to 2025 |
| <b>Multithreaded Path-Tracing Engine</b>                                                                                         | 2023         |
| <b>Infected Utopia Terminal RPG</b> <a href="#">GitHub</a><br><i>Game Designer, Narrative and Combat Systems (4-person team)</i> | 2023         |

TECHNICAL SKILLS

|                                                                                                   |
|---------------------------------------------------------------------------------------------------|
| <b>Agents</b> Agentic/loop programming/engineering, agent harnesses (Pi, Claude Code, Codex), MCP |
| <b>Programming &amp; ML</b> Python, C++, C, C#, Java; PyTorch (FSDP), DeepSpeed, Ray, vLLM        |
| <b>Robotics</b> ROS, LeRobot, ALOHA (Piper, ARX), Panda                                           |
| <b>Hardware Prototyping</b> 3D modeling, 3D printing, soldering                                   |
| <b>Languages</b> Mandarin Chinese, English (fluent)                                               |